MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular EXAMINATION October 2025
Third SEMESTER
Discipline Specific Core Course (DSC)-MG3DSCBOT201
Discipline Specific Elective (DSE)-MG3DSEBBT200 – MYCOLOGY AND PLANT PATHOLOGY
(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Part A
Short answer type questions Answer any 10 questions Each question carries 1 mark [1x10 = 10]

1. What is the function of the urediniospores in *Puccinia*? [K] / [CO1]

   In the *Puccinia* life cycle, the primary function of urediniospores is to serve as the repeating, asexual stage that enables the rapid spread of rust disease during the growing season **(Liberal valuation).**

2. What is the main method of asexual reproduction in *Xylaria*? [K] / [CO1]

   Conidia

3. Whether members of Myxomycotina are Eucarpic or Holocarpic? [U] / [CO1]

   Holocarpic

4. As per Ainsworth's fungal classification, which are the classes under the division Myxomycota. [K] / [CO1]

   1. Class. Acrasiomycetes (cellular slime molds), 2. Class. Hydromyxomycetes (net slime molds), 3. Class. Myxomycetes (true slime molds), 4. Class. Plasmodiophoromycetes (endo- parasitic slime molds). **(Liberal valuation).**

5. Name a fruticose lichen. [K] / [CO1]

   Usnea

6. Name one lichen used in perfumery. [K] / [CO2]

   Evernia prunastri or some *Usnea* sp or any other species **(Liberal valuation).**

7. Define the term Parasite? [K] / [CO3]

   Parasites are organisms that live in, on or with another organism (host). They feed, grow or multiply in a way that harms their host.

8. Give any one mode of transmission of plant pathogens [U] / [CO3]

   Wind (Anemochory), Water (Hydrochory), Animal (Zoochory), Human (Anthropochory) (Any one)

9. Distinguish between hyperplasia and hypoplasia. [U] / [CO3]

   Hyperplasia: An increase in the number of cells in an organ or tissue. The cells themselves are normal. Hypoplasia: The incomplete development or underdevelopment of a tissue or organ. **(Liberal valuation).**

10. What is Leaf mosaic disease of Tapioca? [U] / [CO3, CO4]

    Leaf mosaic disease in tapioca is a viral infection that causes mosaic-like yellow or light-green patterns on leaves, leading to stunted growth and reduced tuber size.

11 Mention any two-disease resistant variety of rice? [K] / [CO4]

KAU Aadhya (MO 24), KAU Punya (MO25), Pusa Basmati and Pusa Basmati (Any two) **(Liberal valuation).**

12. Mention an example of a therapeutic measure in plant disease management. [K] / [CO4]

Antibiotics, fungicides, , heat therapy, surgical REMOVAL (Any one) **(Liberal valuation).**

Part B

Short Essay type questions, Answer any 6 questions, Each question carries 5 marks [5x6 = 30]

13. Describe the thallus of *Agaricus* and its characteristic features, including the role of the mycelium and the formation of basidiocarps. [K] / [CO1]

- The mycelium of basidiomycetous fungi consists of well developed, septate hyphae. The cell wall is made of chitin and glucans, cellulose is absent.
- Thallus of *Agaricus* shows primary, secondary, and tertiary mycelium
- Role of mycelium are absorption and reproduction
- Formation of basidiocarp
- Under moist and nutrient-rich conditions, the dikaryotic mycelium aggregates to form small hyphal knots.
- These develop into primordia or buttons, which later grow into mature basidiocarps (mushrooms).

1. The basidiocarp differentiates into:, Pileus (cap), Stipe (stalk), 2.Gills (lamellae) beneath the pileus, which bear the basidia where karyogamy, meiosis, and basidiospore formation occur

14. Explain the role of the conidia in the life cycle of *Fusarium*, including their formation, dispersal, and germination. [K] / [CO1]

- The members of the genus *Fusarium* typically produce two types of conidia that are termed macroconidia and microconidia based on the size.
- Macroconidia are long, multiseptate, crescent or fusiform in shape, and are generally borne in sporodochia but may also occur separately.
- Microconidia are very small conidia produced from the tips of simple or branched conidiophores, which are indistinguishable from hyphae
- The role of the conidia in the life cycle of *Fusarium* is dispersal. Conidia are non- motile and are dispersed by wind, rain splash, irrigation water, insects, and farm tools.
- Germination of Conidia
- Under favourable conditions (moisture, warmth, and nutrient availability), conidia germinate by producing germ tubes.

15. Differentiate between zoosporangia and conidia in *Albugo*. [U] / [CO1]

Zoosporanigia

In the zoosporangia, the protoplast divides mitotically during germination, giving rise to characteristic zoospores, which after a period of motility round up and germinate to give rise to sporothalli, thus repeating the diploid generation. Represents the typical asexual reproduction.

Conidia

Modified sporangia that act directly as conidia. Represents a modified asexual reproduction under stress or dry conditions in *Albugo.* Direct germination – sporangium germinates by producing a germ tube directly

16. Differentiate between upper cortex and lower cortex of *Parmelia* thallus. [U] / [CO1]

Upper Cortex

- Made of tightly packed fungal hyphae, forming a dense, thick-walled protective skin. May contain specialized pores for gas exchange called pseudocyphellae, as well as vegetative reproductive structures like isidia and soredia. It shields the internal algal layer from dehydration, ultraviolet radiation, and physical damage.

Lower Cortex
Composed of tightly packed, dark-coloured fungal hypha.
Rhizines, which are fungal hyphae that extend downward to attach the lichen to its substrate.

Anchorage and Absorption. Its rhizines act as holdfasts, attaching the lichen to rocks or trees. They also play a role in absorbing water and minerals. **(Liberal valuation).**

17. Explain mycorrhiza its types, structure, and significance. [U] / [CO2]

- A symbiotic relationship between the hyphae of some soil-borne fungi and roots of the other plants are called mycorrhiza. Mycorrhizae provide extra nutrients to the plant.
- Ectomycorrhizae (or Ectotrophic mycorrhizae) - The ectomycorrhizae are characterized by the presence of an external pseudoparenchymatous sheath, called mantle on the terminal nutrient absorbing rootlets
- Endomycorrhizae: The fungal partner grows mainly inside the roots, penetrating the outer cortical cells of the plant root. Only a small portion of the fungal component lies externally as a loose mass of hyphae in soil.
- Ectendomycorrhizae In certain cases, mycorrhizae share characteristics of both ecto- and endo-mycorrhizae..
- Structure and Significance (Liberal valuation)

18. What are phytoalexins? Give examples [K] / [CO3]

- Phytoalexins are antimicrobial compounds produced by plants in response to stress, such as microbial infections or physical injury, as a defense mechanism. Examples: resveratrol from grapevines, pisatin from peas, glyceollin from soybeans, and isoflavonoids from legumes.

19. List any five advantages of biological control. [K] / [CO4]

Environmental Safety; Specific: Biological control agents are often highly specific to the pest; **No Resistance to Approach**: Natural predators or parasites are far less likely to allow pests to become resistant than the use of synthetic chemical products; **Cost Effective**; **Less Risk of Harm. Improve biodiversity, Target pest etc.** or any relevant advantages **(Any 5 points).**

20**.** Describe the thallus structure and vegetative reproduction of *Saccharomyces* [U] / [CO1]

- Thallus Structure

All yeasts, including the species of *Saccharomyces*, are unicellular. The vegetative cells are generally diploid. Even though the cells appear colorless individually, on artificial solid media, colonies appear as white, cream coloured or tinged with brownish pigment. The cells contain a large membrane bound vacuole and a small nucleus. Other inclusions are mitochondria, storage granules, endoplasmic reticulum

- Vegetative Reproduction

In yeasts asexual reproduction is by budding.The cells of *Saccharomyces* can give rise to buds from all over the surface. The point of origin of buds is at the poles of the yeast cell. Possibly it is because the internal fluid pressure in an ellipsoidal cell is exerted maximally against the cell wall at the point of maximum curvature. The cell wall then bulges and stretches out to form the bud.

Part C

Essay type questions **Answer any 1 question,** Question carries 10 marks; (1x10: 10)

21. Describe in detail the vegetative body and asexual reproduction in *Rhizopus*. [U] / [CO1]

The vegetative body in *Rhizopus*

- The vegetative structure is mycelial; hyphae are branched and coenocytic.
- The characteristic feature is the presence of rhizoids at the base of sporangiophores and the stoloniferous habit of mycelium.
- Aerial hyphae when touch the substratum bears rhizoids and sporangiophores.
- The mycelium is differentiated into rhizoids, stolons and sporangiophores.
- The rhizoids are repeatedly branched, penetrate the substratum and anchor the mycelium.
- The stolons are aerial and arching, running horizontally and parallel to the substratum. At some points of contact with the substrate, the stolons develop tufts of rhizoids and enter into the substratum.
- The sporangiophores grow upwards from the stolons exactly opposite the rhizoids, singly or in clusters.

The asexual reproduction in *Rhizopus*

- It occurs by means of aplanospores produced in the sporangia. The sporangiophores develop opposite to the rhizoids
- Sporangia develop terminally on sporangiophores which may be simple or branched.
- The apex of aerial hypha swells and cytoplasmic mass along with nuclei move into this part.
- The swollen part enlarges and develops into a large, globose structure, and it is the young sporangium.
- At the time of spore release, the sporangial wall cracks into a number of small fragments, the columella collapses and the spores are quickly blown away by wind.

22. Discuss the role of fungi in the production of alcohol, beer, wine, and fermented food products. [U] / [CO2]

Alcohol*: Saccharomyces*, the yeast ferments sugars (glucose, sucrose, maltose) into ethanol and carbon dioxide under anaerobic $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$ conditions.

Beer*: Saccharomyces cerevisiae* (top-fermenting yeast) or Saccharomyces carlsbergensis (bottom-fermenting yeast)

- Malting: Barley grains are germinated to produce enzymes that convert starch into maltose.
- Mashing: The malt is mixed with water to form wort (a sugar-rich liquid).
- Fermentation: Yeast ferments the sugars in the wort into ethanol and CO_2, producing beer

Wine

- Natural or inoculated yeast ferments the sugars (mainly glucose and fructose) in grape juice to ethanol and CO_2. The alcohol dissolves the metabolites forming the wine.
- The type of yeast, sugar content, and fermentation conditions influence the flavour, aroma, and alcohol content.
  Secondary fermentation (in sparkling wines) adds carbonation

Fermented food products

- Bread: *Saccharomyces cerevisiae* ferments sugars in dough, releasing CO_2 that causes the dough to rise.
- Cheese: Molds such as *Penicillium roquefortii* (blue cheese) and *P.camemberti* (Camembert) develop characteristic flavours and textures.
- Soy sauce and miso: *Aspergillus oryzae* breaks down starches and proteins during fermentation.

Rhizopus oligosporus ferments soybeans into a protein-rich, digestible product. **(Liberal valuation).**